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Palaeomagnetism of the Cordillera de la Costa, northern Chile: evidence for local forearc rotation

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Palaeomagnetic studies of Lower Jurassic to Lower Cretaceous volcanic, plutonic and sedimentary rocks from the Cordillera de la Costa of northern Chile, indicate that there has been no significant latitudinal movement of northern Chile since early Cretaceous times. A palaeomagnetic pole for the Cordillera de la Costa (63°S, 187°E; $\alpha_{95} = 7.7°$) was derived from combining the results of three studied units, the La Negra Formation (Jurassic volcanics); Jurassic/Lower Cretaceous granodioritic plutons; and the El Way Formation (Hauterivian marine carbonates) together with the Coloso Formation (Lower Cretaceous red beds). Comparison with a Cretaceous reference pole for the stable shield area of South America (87°S, 235°E; $\alpha_{95} = 2.4°$) indicates that $24 \pm 7°$ of clockwise rotation has taken place.

Possible mechanisms for the rotation of the Andean forearc in this area have been examined in relation to the kinetics of the subducting Nazca and over-riding South American plates. The Cordillera de la Costa has been subjected to both compressional and extensional tectonics since the Lower Cretaceous. Major eastwardly directed thrusting in mid-Cretaceous and early Tertiary times, possibly associated with a period of oblique subduction of the Nazca plate, contributed to the accretion of the Cordillera to the main forearc; rotation may have occurred during this process. In Oligo-Miocene and later Tertiary times, extensional faulting (possibly associated with roll-back of the Nazca plate) led to the reactivation of the Atacama fault zone and the formation of oblique extension faults in the Cordillera de la Costa. Structural rotation associated with this phase of extension could equally explain the observed palaeomagnetic results.

1. Introduction

In recent years the influence of the angle of subduction and rate of convergence in determining the tectonic framework of forearc regions has been recognised (e.g. [1]), and palaeomagnetism has had an influential role in determining whether rotation and/or latitudinal displacements of discrete structural blocks has taken place. The technique has been particularly well exploited at convergent plate margins notably: the North American Cordillera, Southeast Asia, New Zealand, and the Eastern Mediterranean (e.g. [2–4]). Here, we report palaeomagnetic results from the Andean forearc of northern Chile, and discuss their implications for the tectonic evolution of the Andean forearc.

Palaeomagnetic studies of late Palaeozoic and Mesozoic rocks from the Central Andean forearc have yielded both concordant and discordant results when compared with the apparent polar wander path (APWP) for South America [5–15]. The results fall into two geographically distinct groups: north of 19°S anticlockwise rotations have been recorded [6–9], whereas to the south of 19°S, clockwise rotations and non-rotational results have been documented [9–15]. Also, with the exception of one Triassic Formation from central Chile [14] no latitudinal displacement has been recorded from the studied sequences. This is in contrast to the North American Cordillera which is mainly comprised of allochthonous terranes that characteristically show a significant latitudinal displacement (e.g. [2]), although, it should be noted that the magnetization of some of the studied units from the Central Andes may not be primary in origin. Overall it appears that allochthonous terranes have had only a minor role in Central

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Andean orogenesis, a point noted by other workers [16,17], although, the presence of large Precambrian/Late Palaeozoic massifs within the Cordillera de la Costa of South America has yet to be satisfactorally explained [14].

Here, we report palaeomagnetic results from Jurassic/Lower Cretaceous volcanics, plutonics and sediments from the Cordillera de la Costa of northern Chile. The study supplements previous work [11], and seeks to establish whether rotation

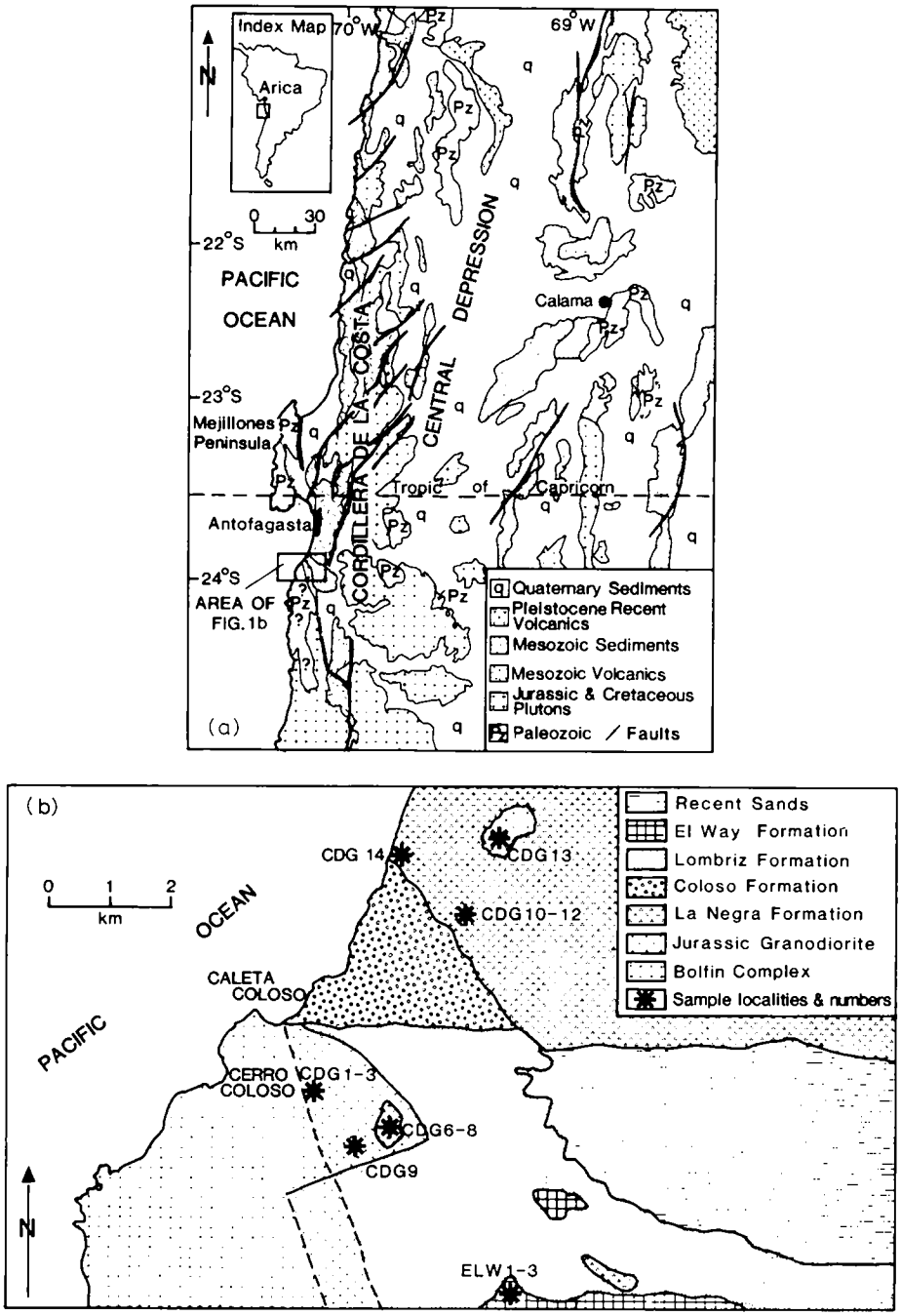


Fig. 1. (a) Geological map of the Cordillera de la Costa, northern Chile, with an insert of the study area (after [44]). (b) Geological map of the study area showing palaeomagnetic sampling localities [45].

of the forearc has occurred in this area and if so, to identify the age, amount, origin and possible method of rotation.

2. Geology

The Cordillera de la Costa, Antofagasta Province, comprises rocks ranging in age from Precambrian [18] to Quaternary [19] and is bounded to the east by the Atacama fault zone (Fig. 1a). Exposure is generally good, and it is dominated by the apparently 10 km thick, volcanic La Negra Formation [20], and associated acid to intermediate Jurassic and limited Cretaceous plutons [19]. Further south in the Chanaral district extensive Jurassic gabbros occur within the Cordillera de la Costa [21]. In Antofagasta Province, the plutons range in age from 159 to 134 Ma [19,22] and intrude the mid-Jurassic La Negra Formation [22]. The La Negra Formation and plutons have been interpreted as representing the relicts of a volcanic island arc [23–25]. Alternatively, the La Negra Formation may have been erupted through fissures in an ensialic back-arc basin [22]. Lower Tertiary volcanics have been recorded from the Antofagasta area [19], however, as there are no age determinations and the stratigraphic position of the rocks is unclear, their age must remain in doubt.

The La Negra Formation and granodiorites are overlain (with a low-angle unconformity) by the Lower Cretaceous continental sediments of the Coloso and Lombriz Formations [19]. This age was assigned to the Coloso and Lombriz Formations as they are conformably overlain by the shallow marine sandstones and limestones of the El Way Formation which contain Hauterivian macrofossils [26]. Along the coastal margin, Tertiary and Quaternary aeolian and shallow marine sandstones and gravels unconformably overlie all older rocks [19].

Extension faults with a NE-SW trend occur throughout the Cordillera de la Costa [27]. They are bounded to the east by the Atacama fault zone, a major basement lineament which was reactivated in Tertiary times [28].

3. Sampling and laboratory methods

Hand samples and drill cores were collected from the La Negra Formation, Jurassic granodioritic plutons and the El Way Formation (Fig. 1b). The samples were oriented using both magnetic and sun compasses. The maximum observed difference between the two methods was $\pm 2°$. From sites drilled in the field and block samples drilled in the laboratory, 4–6 oriented specimen cores were obtained (2.54 cm in diameter and height). One polished thin section was made from each site, which together with the isothermal remanent magnetization (IRM) technique [29] was used to determine the magnetic mineralogy of selected specimens.

The initial natural remanent magnetization (NRM) was measured using a Digico balanced fluxgate magnetometer. Specimens were treated using a step-wise thermal demagnetization technique. All palaeomagnetic results have been tectonically corrected for the local dip of the La Negra and Coloso Formations; both the formations dip approximately 20° to the southwest. Although the La Negra Formation may not have been deposited horizontally, it is likely that the overlying marine sediments of the El Way Formation were, and so it may be assumed that the whole of the area underwent post-Hauterivian tilting to the southwest. An *F*-test [30] was applied to all samples, but the results proved inconclusive (Table 1).

Thermal demagnetization was carried out using a non-inductive furnace and Helmholtz coil system in which the field is continuously monitored. Ambient magnetic fields during heating can be controlled to less than 2γ using this method. NRM measurements after each heating cycle were made using a two-axis cryogenic magnetometer for weakly magnetized specimens calibrated with a Digico balanced fluxgate magnetometer which was used for more strongly magnetized specimens.

All the palaeomagnetic directions and pole positions reported here are based on mean directions of discrete, isolated components of magnetization. No single-step heatings or bulk demagnetizations have been used in the analysis.

4. Palaeomagnetic results

4.1. The La Negra Formation

Seven sites ($n = 35$) in andesitic volcanic rocks from the La Negra Formation were analysed. Initial intensities of NRM varied between 15.4 and

TABLE 1

Components of magnetization extracted from the studied units of the Cordillera de la Costa using the LSF method, and the respective temperature ranges

Site	N	R	K	CSD	α_{95}	DEC	INC	Range (°C)
La Negra Formation (all data tilt corrected, tilt correction = 120/20 SW)								
CDG6	4	3.947	117 (118)	7.5 (7.5)	8.5 (8.5)	16 (328)	−22 (−52)	600–680
CDG8	6	5.978	222 (231)	5.4 (5.3)	4.5 (4.5)	24 (22)	38 (−58)	500–680
CDG12	3	2.999	3365 (1891)	1.4 (1.9)	2.1 (2.8)	32 (77)	−38 (−38)	400–500
CDG14	6	5.750	20 (20)	18.1 (18)	15.1 (15.1)	29 (29)	−32 (−52)	0–500
CDG7	6	5.967	156 (162)	6.5 (6.4)	5.4 (5.3)	189 (207)	+48 (+64)	500–680
CDG8	5	4.940	67 (69)	9.9 (9.8)	9.4 (9.3)	223 (242)	+58 (+77)	0–150
CDG12	3	2.994	356 (361)	4.3 (4.3)	6.5 (6.5)	236 (247)	+37 (+54)	0–150
CDG10*	4	3.964	83 (81)	8.9 (9)	10.1 (10.3)	199 (205)	−75 (−56)	0–550
CDG11*	4	3.992	385 (207)	4.1 (5.6)	4.7 (6.4)	271 (241)	−75 (−60)	0–550
Jurassic granodiorites (all data tilt corrected, tilt correction = 120/20 SW)								
CDG1	5	4.802	20 (20)	18.0 (18.3)	17.4 (17.7)	203 (201)	+20 (+40)	620–680
CDG3	5	4.875	32 (32)	14.3 (14.3)	13.7 (13.7)	226 (231)	+28 (+47)	0–680
CDG13	5	4.674	12 (13)	23.1 (22.9)	21.8 (21.5)	207 (205)	+36 (+55)	300–680
El Way Formation (all data tilt corrected, tilt correction = 105/22 SW)								
ELW1	6	5.930	72 (64)	9.6 (10)	8.0 (8.4)	10 (8)	−29 (−52)	0–580
ELW2	6	5.908	54 (58)	11 (11)	9.2 (9.1)	25 (28)	33 (54)	0–500
ELW3	6	5.793	24 (24)	16.5 (16.7)	13.9 (14.1)	24 (28)	−32 (−54)	0–500

CDG1, 3, 13 = Jurassic granodiorite; CDG6, 7, 8, 12, 14 = La Negra Formation (normal and reversed); ELW1, 2, 3 = El Way Formation.

N = Number of specimens per site; R = resultant vector [46]; K = estimated precison parameter [46]; CSD = circular standard deviation; α_{95} = angular radius of 95% cone of confidence about the observed mean; DEC = declination of the isolated component in degrees; INC = inclination of the isolated component in degrees with north seeking directions down taken as positive.

Figures in parentheses refer to structurally uncorrected data for the extracted components.

* Not included in pole position calculation.

205.4×10^{-3} A m^{-1}. Three groups were determined from variations in site behaviour during demagnetization, and the extracted components are presented in Table 1.

Group 1 (CDG7, 8, 12, 14): Both CDG8 and 12 contain reversed components relative to the present geomagnetic direction which are removed between 0 and 150°C; individual specimens of CDG7 and 14 contain this component. This component is, however, poorly defined ($\alpha_{95} > 25°$) and has therefore been excluded from Table 1. The component is characterised by a sharp increase in intensity up to 150°C after which half the NRM is lost between 300° and 500°C, as shown by the normalised intensity decay curves (Fig. 2a). A component of normal polarity predominates between 150 and 500–580°C (Table 1). Above 500 580°C a reversed component is isolated and is coincident with a marked unblocking temperature (Fig. 2a). Fig. 2b shows the large directional changes which occur from an initial position in the NE quadrant, to a steep southerly direction as a lower unblocking temperature component is removed. The Zijderveld orthogonal diagrams allow a better definition of the normal and reversed components.

Group 2 (CDG10, 11): This group contains one dominant component, interpreted as intermediate in nature, since it lies directly between the normal and reversed components (Table 1). All analytical techniques indicate a one component system; straight line segments on Zijderveld orthogonal diagrams, stable positions on stereographic projections, similar isolated components over all temperature ranges and normalised intensity decay curves with no marked unblocking temperatures and a gradual decrease in NRM (Fig. 2a).

Group 3 (CDG6): CDG6 displays significantly different behaviour to the samples in the other groups. A rapid decrease in NRM, with 80% lost by 150°C, is shown by the normalised intensity decay curve (Fig. 2a), corresponding to the re-

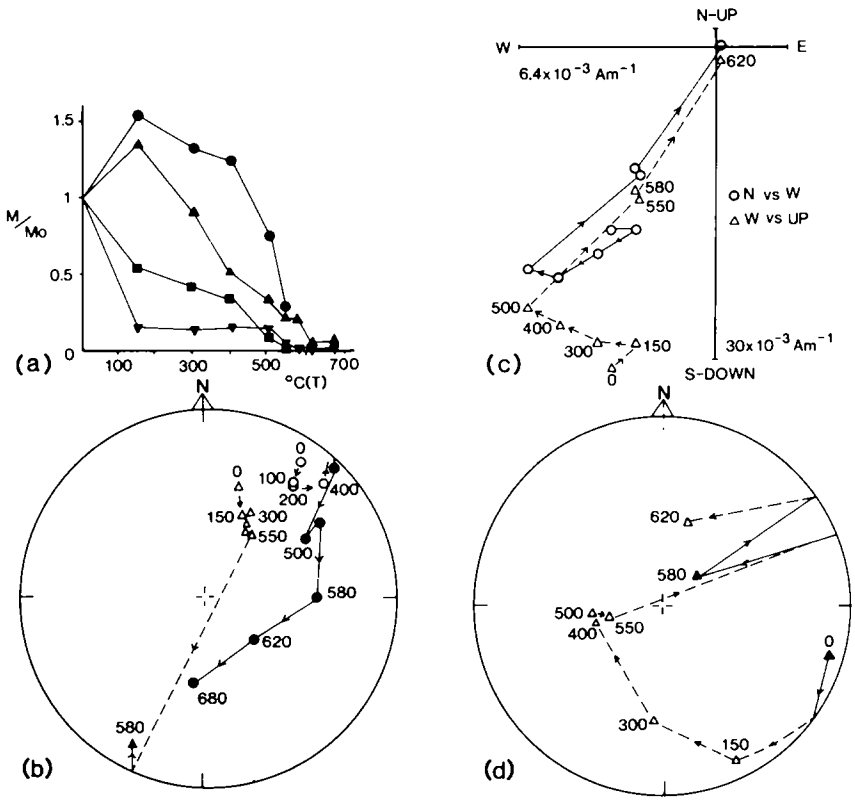


Fig. 2. Palaeomagnetic analysis of the La Negra Formation. (a) Decay curves for CDG6 (▼), CDG8 (▲), CDG11 (■) and CDG12 (●). (b) Stereographic projection of demagnetization behaviour for CDG8 (▲) and CDG14 (●). (c) Zijderveld diagram for CDG7, contains three components, all with sharp breaks in their representative unblocking temperature spectra (10–150 °C = low-temperature component, 150–500 °C = normal component, 500–620 °C = reversed component). (d) Stereographic projection of demagnetization behaviour for CDG6. Open symbols = upward inclination, closed symbols = downward inclination.

moval of a low unblocking temperature component. The group also contains a 150–500 °C component interpreted as intermediate in nature characterised by a marked unblocking temperature at 500 °C and a 580–680 °C normal component. The 150–500 °C component is thought to be transitional, as both stereographic projections and orthogonal plots indicate continuous movement of the isolated components throughout demagnetization, finishing in the NE quadrant (Fig. 2d).

IRM acquisition curves and reflected light microscopy indicate that magnetite and haematite are present as discrete grains in all three groups. However, group 3 also contains cross-cutting veins of haematite, thought to be secondary in origin. It should be noted that high-temperature components may have been present in CDG10, 11 and 12, but due to disintegration of the specimens between 500 and 580 °C, measurement was impossible.

4.2. Jurassic granodioritic plutons

Two granodioritic plutons were sampled from the Cordillera de la Costa (Fig. 1b) which yielded 28 specimen cores from 5 sites. Initial intensities of NRM vary between 7.4 and 488.1 $\times 10^{-3}$ A m^{-1}. Of the five sites, three show similar behaviour during demagnetization (CDG1, 3 and 13), whereas two (CDG2 and 9), which have been altered and mineralized, exhibit different behaviour.

Normalised intensity decay curves reveal a marked unblocking temperature (at 500–550 °C), coincident with a polarity change for CDG1, 3, 9 and 13 (Fig. 3a). Normal components dominate at low temperatures and reversed components at high

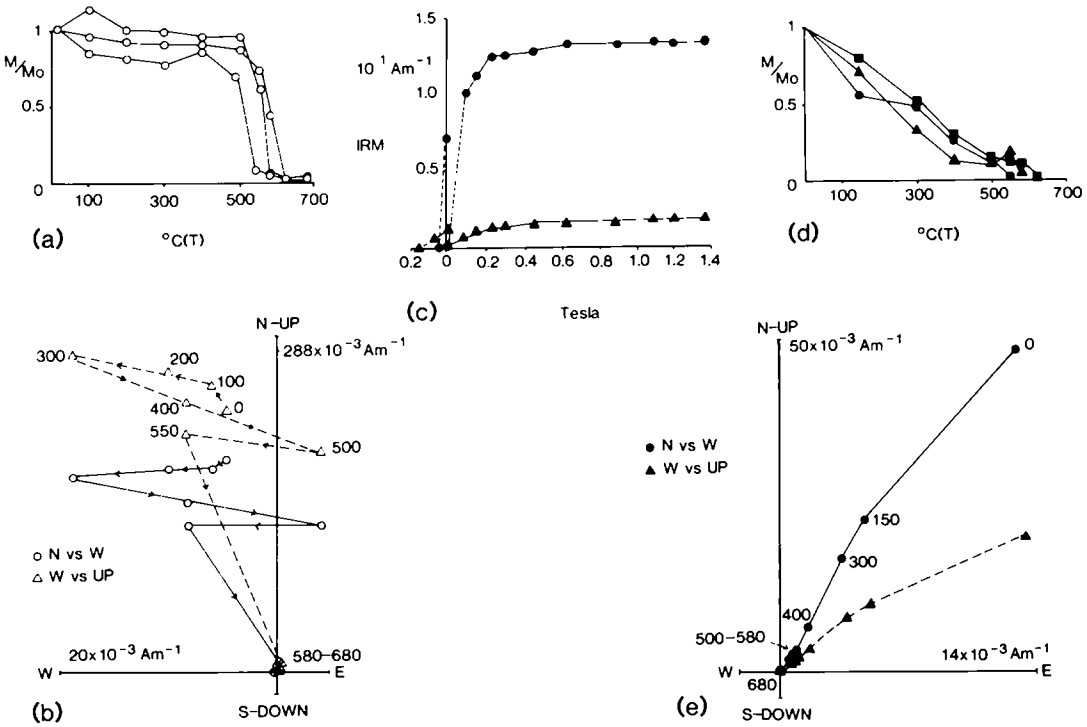


Fig. 3. Palaeomagnetic analysis of Jurassic granodiorites: (a) decay curves for CDG1, 2 and 3; (b) Zijderveld diagram for CDG2, showing 3 linear components; (c) IRM acquisition curves for ELW1 (●) and ELW2 (▲). Palaeomagnetic analysis of the El Way Formation: (d) decay curves for ELW1 (■), ELW2 (●) and ELW3 (▲); (e) Zijderveld diagram for ELW1. Other details as Fig. 2.

temperatures (Fig. 3b). The normal components and components isolated from CDG2 and 9 are, however, poorly defined ($\alpha_{95} > 20°$) and have been excluded from Table 1. IRM acquisition curves indicate that the plutons contain both magnetite and haematite.

4.3. The El Way Formation

Two limestone beds (ELW2, 3) and a calcareous sandstone bed (ELW1), were drilled in the field. Initial intensities of NRM varied from 1.91 to 157×10^{-3} A m^{-1} with the stronger intensities being characteristic of the sandstone unit. Polished section microscopy and IRM acquisition curves (Fig. 3c) indicate that the magnetic carrier is magnetite. The greater amount of magnetite in ELW1 (sandstone) than ELW2 and 3 (limestone) is probably responsible for the large variation in the initial intensity of NRM between the sandstones and limestones and strongly suggests that the magnetite is detrital in origin.

Examination of the normalised intensity decay curves, reveals similar behaviour for all three sites (Fig. 3d). They show a steady drop in intensity with a marked unblocking temperature at 580°C for ELW1 and at 500–550°C for ELW2 and 3. Analysis of the stereographic projections allows a correlation to be made with the decay curve observations. Specimens from all three sites have a stable normal component which only moves when the respective unblocking temperatures are reached (Fig. 3d). Inspection of the Zijderveld diagram reveals the same behaviour, straight one-component demagnetization until the respective unblocking temperatures are reached (Fig. 3e). Components below the unblocking temperature have a normal direction of magnetization.

4.4. Interpretation and age of magnetization

La Negra Formation: The magnetization history of the formation is complex, containing antiparallel, normal and reversed components, consequently it may not record an original lower/middle Jurassic age of extrusion. The complex

behaviour is likely to be related to the emplacement of the coastal batholith and remagnetization associated with hydrothermal fluids.

Jurassic granodiorites: Antiparallel normal and reversed components of magnetization are present in the granodiorites, indicating a complex history of component acquisition which may not reflect the original age of intrusion. Also, as two of the sites contain copper mineralization, late-stage hydrothermal fluids are likely to have influenced the magnetization history.

El Way Formation: A single normal component of magnetization carried by detrital magnetite is present in the El Way Formation. Petrographic evidence indicates that primary diagenetic textures are present, precluding a metamorphic or hydrothermal event. Also, no petrographic evidence of authigenic magnetite was found. Consequently, the magnetization of the El Way Formation is considered to be a post-depositional or depositional remanent magnetization (PDRM/DRM) of Hauterivian age.

As the magnetization history of both the La Negra Formation and the granodiorites probably reflects the last magmatic event in the Coloso area of the Cordillera de la Costa, age constraints on the timing of igneous activity will be considered. Radiometric dates from the plutons and lavas of the Cordillera de la Costa range from 186 to 134 Ma [18,19,22]. Possible Lower Tertiary activity has been recorded from the Central Depression 150 km to the southeast of Antofagasta [19], but not from the Cordillera de la Costa. Consequently, the La Negra Formation and granodiorites may record components of magnetization as young as Lower Cretaceous, but are unlikely to contain younger components.

The similarity between the isolated components from the studied units and the Lower Cretaceous Coloso Formation (Fig. 4a), may indicate that the magnetization of the studied units is Jurassic to Lower Cretaceous in age. Particularly as the magnetization of the Coloso Formation is considered to be PDRM acquired rapidly after deposition [11], and diagenetic studies indicate a low temperature $< 150\,^{\circ}C$ mineral assemblage with no evidence of hydrothermal alteration [31,32]. However, when considering palaeomagnetic data from Triassic and Mesozoic rocks of the Central Andes, one must be aware of the general post-Carboniferous stability of the South American plate with respect to the geomagnetic dipole [13]. Consequently, component analysis from South American Mesozoic rocks must be viewed with caution,

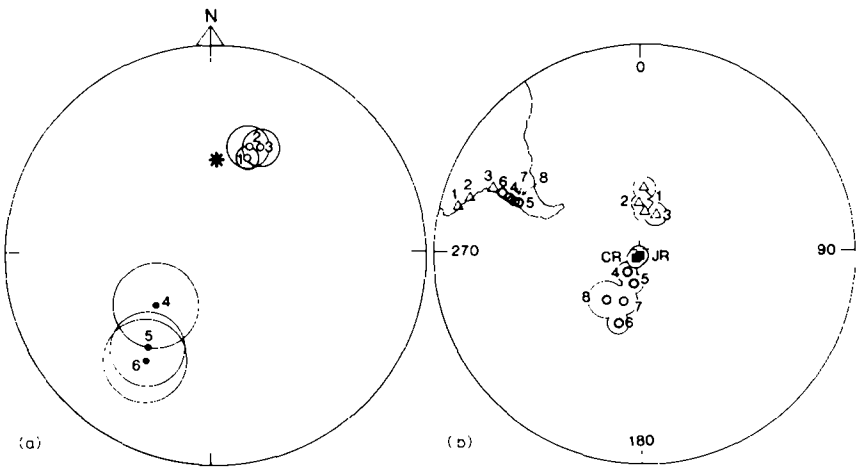


Fig. 4. (a) Equal-angle stereographic projection of extracted components (Table 2). *1* = Coloso Formation (normal) [11], *2* = El Way Formation, *3* = La Negra Formation (normal), *4* = La Negra Formation (reversed), *5* = granodiorites, *6* = Coloso Formation (reversed) [11]. * = present axial dipole field. (b) Equal-angle stereographic projection of Mesozoic palaeomagnetic poles derived from the Andean forearc, compared to the Upper Jurassic (*JR*) and Cretaceous (*CR*) reference poles derived from the stable shield area of South America [10]. Triangles show poles rotated anticlockwise, circles show poles rotated clockwise with respect to the reference poles as viewed from South America. References: *1* = [6]; *2* = [8]; *3* = [7]; *4* = [12]; *5* = [9]; *6* = this study; *7*, *8* = [13]. Arrows indicate the sense of rotation.

as the time lapse between primary and any subsequent component acquisition is difficult to resolve.

5. Discussion

A major problem with Cretaceous palaeomagnetic results from South America, is the scattered distribution of poles from stable South America [9,10] making reference pole selection difficult. This could be attributed to the following reasons: (1) that the distribution represents genuine APW [33], (2) that intra-cratonic deformation has affected some of the sampled sites, or (3) true non-secular variation has taken place. In order to minimise the effect of the scattered distribution of Cretaceous cratonic poles, the reference pole used here is that calculated by Beck [10] for the Cretaceous from seven poles (ranging from 126 to 75 Ma) located in the centre of the scattered distribution. For further details and discussion concerning this reference pole see [10].

Comparison of the pole derived from the Cordillera de la Costa (63°S, 187°E; Table 2), with respect to the Cretaceous reference pole (85.5°S, 253°E), indicates that substantial clockwise rotation ($24 \pm 7°$) has taken place. Rotation must be post-Hauterivian in age. The error limit has been calculated assuming that the age of magnetization of the sampled rocks from the Cordillera de la Costa is Cretaceous. However, even if the age of magnetization is Tertiary, the Tertiary reference pole for South America is so close to the present-day axial dipole [34,35] to make the difference in amount of rotation due to later remagnetization negligible.

Clockwise rotation of the Andean forearc has also been recorded from remagnetized Permo-Triassic volcanics and marine sediments from central Chile [13], Cretaceous ash flow tuffs from the Santiago area [9] and Cretaceous volcanics from the La Serena area [12]. Non-rotational palaeomagnetic results have also been obtained from remagnetized Cretaceous intrusions (similar to the Coloso granodiorites) south of La Serena [15], and late Palaeozoic volcanics and Triassic sediments from the Precordillera of northern Chile [14]. All the above results, including those from the Coloso area, are located south of 23°S.

In contrast to the above results, anticlockwise rotation has been recorded north of 19°S from Jurassic lavas and dykes in the Arica area [7]. Further north in Peru, anticlockwise rotation has been found in Cretaceous volcanics from southern and central Peru [5,6,8]. Fig. 4b summarises all good-quality rotational palaeomagnetic data from the Central Andean forearc.

The extent of palaeomagnetically identified rotation (between 6° and 34°S), and the variation in amount and direction (Fig. 4b), indicates that

TABLE 2

Statistics for the isolated components from the units studied (Table 1), and palaeomagnetic pole

Unit		N	(n)	R	K	CSD	α_{95}	DEC	INC	Pole	
										lat.	long.
Jurassic granodiorites		3	(15)	2.945	36.0	13.5	20.8	212	+28	59°S	189°E
La Negra Fm	N	4	(19)	3.958	70.8	9.6	11.0	25	−33	66°S	190°E
	R	3	(14)	2.884	17.2	19.5	26.4	217	+50	57°S	220°E
Coloso Fm	N *		(17)	16.435	28.3	15.2	6.8	21	36	70°S	192°E
	R *		(9)	7.948	7.6	29.4	20.0	217	+31	55°S	195°E
El Way Fm		3	(18)	2.983	117.1	7.5	11.4	20	−32	70°S	184°E
Combined		6		5.936	78.0	9.2	7.7	28.4	−35.2	62.2°S	186°E
Expected dir.								4.5	−35.9		
Rotation				−23.9		±7.4					
Flattening								−0.2	±7.3		

N = number of sites per unit; (n) = number of specimens per site.

N = normal, R = reversed; Expected dir. = the expected declination and inclination of the study area, assuming a Cretaceous dipole magnetic field with a pole at 85.5°S, 253.5°E, $\alpha_{95} = 2.4°$ [10]; Flattening = difference between expected and observed inclination; Rotation = difference between expected and observed direction and 95% confidence limits [2,47]. For explanation of other symbols used see Table 1.

* From Turner et al. [11].

rotation must have resulted from the "localised" movement of discrete structural blocks. Non-rotational palaeomagnetic data also supports this "localised" hypothesis, and with more data may allow some constraints to be placed on the size of the rotating structural blocks. In central and northern Chile rotation has affected areas within the Andean Cordillera up to 200 km from the Pacific Coast [13,32]. This indicates that whatever the mechanism of rotation, in some regions it affects the whole of the Andean forearc.

5.1. Mechanism of rotation

A mechanism for rotation within the Andean forearc should be related to the dominant stress regime at the Nazca–South American convergent plate boundary. Examination of the early Cretaceous/Recent tectonic history of the Andean forearc may, therefore, reveal conditions favourable for block rotation.

Since the early Cretaceous four compressional tectonic phases have been documented from the Central Andes: the mid-Cretaceous Peruvian phase

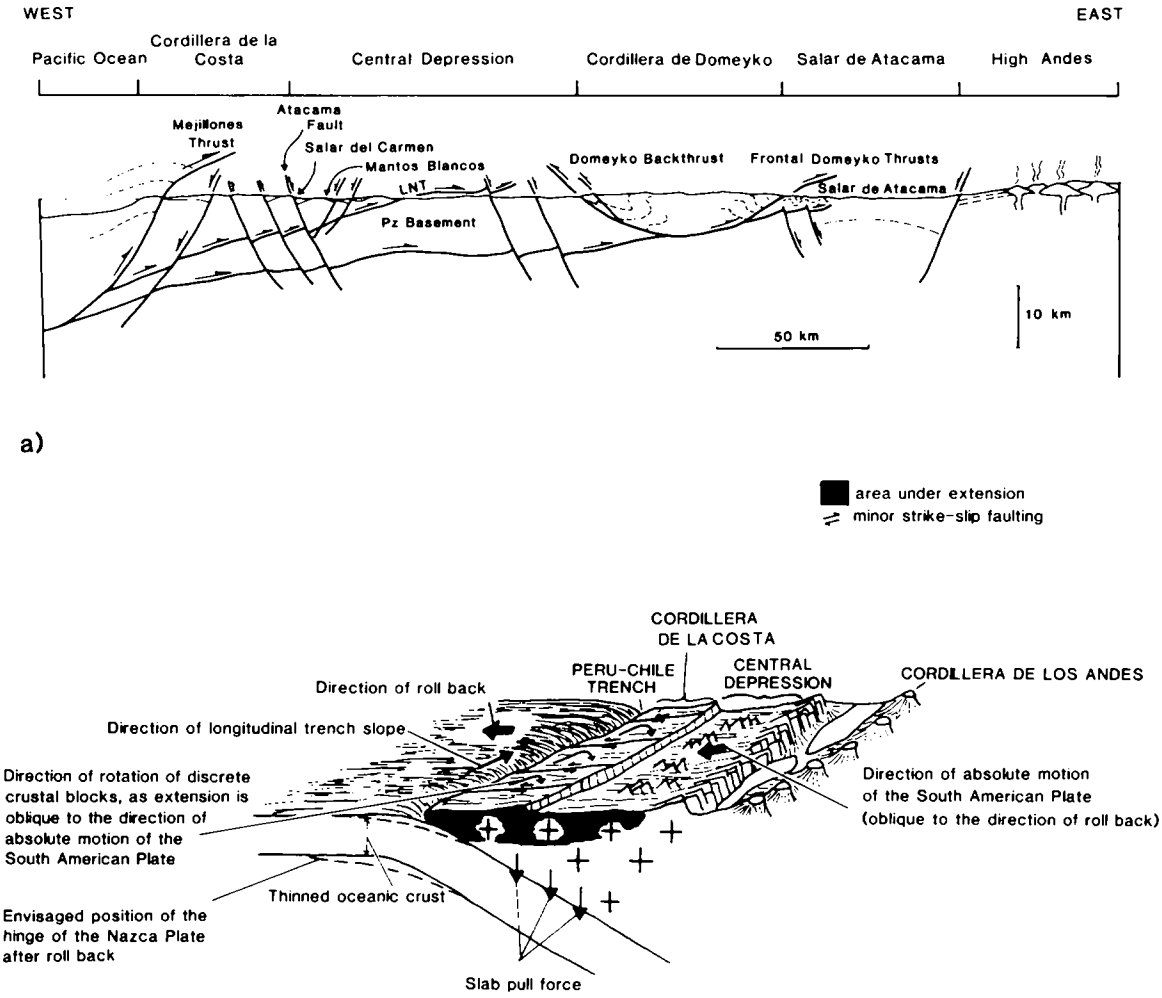


Fig. 5. (a) Schematic structural cross section of the Andean Forearc (23°S), showing the dominance of extension faulting in the Cordillera de la Costa and Central Depression. *LNT* = La Negra Thrust. (b) Schematic diagram of the Andean forearc for northern Chile, showing the effects and processes resulting from the extension of the South American plate at an oblique angle to its direction of absolute motion.

[23], the mid-Eocene Incaic phase [23,36], the mid-Miocene Quechua phase [23,37] and the late Pliocene Diaguita phase [38]. These phases are particularly well documented in southern and central Peru, and have also been recorded from northern Chile ([38], and our own field observations). These compressional events have further been correlated with relatively rapid sea-floor spreading [39], separated by periods of relatively slow and oblique (late Cretaceous/early Tertiary) convergence along the Andean margin [39,40].

Rotation related to the different movement of thrust sheets has been well documented in areas dominated by compressional tectonic regimes, and may be responsible for the rotation of the Andean forearc. It could account for the disparity between Peruvian and Chilean palaeomagnetic results, through the deflection of the sense of compression around a buttress located in the Arica area (a similar suggestion has been proposed by Beck in an oblique shear model [10], see below). Compression-related rotation would also explain the inland rotation. However, the lack of post-early Cretaceous compressional related features in the Cordillera de la Costa, and dominance of extensional tectonic features, such as the Atacama fault and NE–SW oriented extensional faults (Figs. 1, 5a) do not support this hypothesis.

Alternatively, rotation may be related to a period of right-oblique convergence of the Nazca and South American plates during the late Cretaceous/early Tertiary [9]. This is in the correct sense to cause dextral shear and associated clockwise rotation in the Chilean continental margin [9]. However, this is not the correct sense of shear to explain the anticlockwise rotation of the Peruvian continental margin. This model also has difficulty in explaining the extent of forearc rotation [13].

In view of the presence of extensional tectonic features in the Cordillera de la Costa and the associated Central Depression (Fig. 5a), rotation may well be related to an extensional tectonic regime. Extension of the forearc at an oblique angle to the present direction of absolute motion of the South American plate, would result in the rotation of discrete fault-bounded blocks. Extension is thought to result from a combination of subduction roll back [41] and slab pull force [42].

The vectors of absolute motion for the South American plate [43] are slightly oblique to the direction of roll back and associated slab pull (assuming the direction of roll back is directly opposite to the direction of subduction of the Nazca plate). Consequently, the South American plate is subject to an extensional stress regime, applied at an angle to the direction of absolute motion of South America. This could be accommodated through the rotation of discrete crustal blocks in the Andean forearc (Fig. 5b).

This theory helps to explain the disparity between Peruvian and Chilean palaeomagnetic data, as anticlockwise rotation could be accounted for by extension in a more southerly direction than the absolute motion of the Peruvian part of the South American plate. However, the theory is difficult to resolve with inland forearc rotation.

In conclusion, all of the possible models for rotation have major problems associated with them and as yet none of them appear to satisfactorily explain the rotation present in the Andean forearc. It appears that the way forward is to constrain the age of rotation and then compare this time period with the tectonic history of the Nazca and South American plates to determine which tectonic regime was dominant during rotation.

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